

PERSONAL INFORMATION:

Date of birth:	13 Aug 1977
Nationality:	Portuguese
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CONTACT DETAILS:

Email:	santos.c@buas.nl
LinkedIn:	Carlos Pereira Santos on LinkedIn
Pure:	Carlos Pereira Santos on Pure

Professional experience:**Since 2009 Breda University of Applied Sciences****Professor Applied Data Science and Artificial Intelligence** (since Jan 2025)

Carlos leads applied Data Science and AI research at BUas, with a focus on data-driven real-time systems, AI models, game engine technology, and distributed computing. He shapes the university's AI policy, infrastructure, and laboratory strategy, and leads applied AI research initiatives aligned with BUas's strategic vision, including the AI Pioneer Team, IT Server Compute Infrastructure and Game Hub.

Cradle Research Lab Coordinator & Lead Researcher (2015 — 2025)

Carlos founded, built, and led Cradle, BUas's applied research lab focused on Games, Media and AI, growing it from an initial concept to a team of 25 researchers (23.0 FTE) delivering over 50 applied R&D projects across games, XR, AI, virtual production, and serious games.

He was responsible for team design, recruitment, professional development, and budget management, culminating in Cradle becoming the only R&D group at a Dutch University of Applied Sciences to receive an all-excellent accreditation in 2024.

Contributed to the design and development of the Applied Data Science & AI (ADS&AI) BSc programme. Designed, and established key BUas lab facilities: XR Stage (Virtual Production Studio), XR Lab, Motion Capture Studio, Photogrammetry Studio, and AI Makerspace, available to research, education and industry. Founded and organised annually since 2020 the Virtual Production Gathering, a lead conference attracting 200+ industry participants.

Senior Lecturer (2009 — 2015)

Carlos joined BUas as a lecturer responsible for core technical and creative modules across bachelor and master programmes, establishing himself as a key contributor to curriculum development in game technology, AI, and interactive media.

Responsible Lecturer for: Artificial Intelligence, Game Networking, Mobile Development, Network Programming, and Experimental Games. Responsible Teacher for the Technology module of the Master Media Innovation (MMI), which he subsequently transitioned into the Master Game Technology (MGT) programme.

2001-2009 GMV / Skysoft Portugal, SA**Technical Director:** Aeronautics, Security and Defence Department

Carlos led the software design and development of technical solutions for Critical Safety Software like RTMC-OSPF and RTMC-IPF Subsystems of Galileo (the European GPS), promoted by the European Space Agency.

Design authority and lead developer on complex network server component, of Naval radar and communication system, deployed in the islands of Azores, Madeira and Canaries. Lead developer of a propeller balancing software for Socata TB-30 Epsilon airplanes for the Portuguese Air Force to improve the vibration reduction.

Lead Programmer on a 3D Aircraft Simulator, development and integration with a Radar Air Traffic Controller Simulator, which is used mainly for demonstrations.

2001-2003 Setúbal Polytechnic University of Technology

Teaching Assistant in Department of System and Informatics

Carlos was a lecturer responsible for Computer Graphics courses, teaching 2D and 3D algorithms using C/C++ and OpenGL. He prepared all theoretical and laboratory material and lectured on Relational Database Systems using Oracle SQL and web connections and interfaces.

1998-2000 New University of Lisbon – Science and Technologic Faculty

Junior Teaching Assistant in Department of Informatics

Lecturer for the courses of Programming level I, level II and level IV. Lecture Relational Database Systems, using Oracle SQL, PL/SQL, Oracle Forms, and Web connections and interfaces.

Education:

2019 Doctor of Philosophy – Understanding People through Games

Promoted by Prof. Dr. Panos Markopoulos from Eindhoven University of Technology, dissertation entitled: *Understanding People through Games*

2004 Master of Science in Computer Science

Supervised by Prof. Dr. Nuno Correia from New University of Lisbon, dissertation entitled: *Contextual information support for Augmented environments*

2000 Software Engineer and Development (5 year) Degree

from New University of Lisbon – Science and Technology Faculty
Distinction award: *Best Software Engineer Student*

Other Professional Training:

62 hours BKE Examination Qualification. Training on the development and revision of educational assessment material and activities.

236 hours BKO Teacher Qualification. Didactics training on the design, organization, and training of teaching material and activities.

30 hours LEGO® SERIOUS PLAY® Facilitator. Facilitate workshops to support thinking, communication and problem solving within organisations, teams and individuals.

80 hours RTCA 178B – Airborne Software Development and Certification. Training by Enea TekSci, about critical software systems, by the FAA (Federal American Aviation) standards.

20 hours LynxOS 178B – Real Time Systems. Training course of development of real time applications, focus on non-stop systems, processing regulation and recovery policies.

36 hours Team Management and Leadership. Participation on a Workshop in Chicago, over the topics of teamwork, leadership and motivation.

R&D Projects (with over 30 completed projects, covering only impactful & relevant):

- 2025-2029** Principal Investigator of **AVENUE** (<https://avenuecove.eu/>) Cove Erasmus+ that developed five leading ecosystems across Europe to build stronger bridges between education, industry, and innovation in the creative visual sector.
- 2026-2028** Principal Investigator of **VPSN2** (<https://vpstudio.network.com/>) - Virtual Production Studio Network (VPSN) 2.0 builds on VPSN initiative, expanding its scope to meet the evolving needs of Virtual Production education and industry practice. Funded by the Erasmus+ programme to advance accessible, industry-focused, and sustainable VP training for educators and students.
- 2024-2027** Principal Investigator of **KreativEU** (<https://kreativeu.org/>) is a Funded European Alliance with long-term institutional, structural and strategic cooperation on Europe's cultural heritage, Carlos is specifically co-leading the Work package on Sustainability of AI in the European context.
- 2023-2025** Principal Investigator of **VPSN** (<https://vpstudio.network.com/>) - Virtual Production Studio Network (VPSN) is a Erasmus+ project to support institutions to integrate VP in their programs, by developing open research, training and educational materials related to Virtual Production practices for educators and students.
- 2024-2025** Principal Investigator of **AICI** (AI in the Creative Industry) an Erasmus European funding focusing on assess the opportunities and challenges of generative AI, and its implications for the Creative Digital Industry, and the development of a Generative Storyboard AI tool.
- 2023-2024** Principal Investigator & Technical Coordinator of **Procedural Ocean** (<https://cradle.buas.nl/>) is financed by Horizon funding and is a pilot in automatic generating 3D environments in Unreal based on real geo-data concerning offshore energy, biology and environments, to aid the learning, policy- and decision-making process of planners and stakeholders of offshore areas.
- 2021-2023** Project Manager of **PlanV** with Painting Practice, a professional tool to support film and television professionals to iterate on set design, cameras, lighting, and animation, reducing production costs while increasing efficiency, and used on His Dark Materials (BBC/HBO) and Wednesday (Netflix).
- 2023** Technical Manager in "**Bound to the Miraculous**" ([Bound to the Miraculous – cradle.buas.nl](https://cradle.buas.nl/)): Led the creation of a simulated ocean environment in Unreal Engine 5, inspired by Bas Jan Ader's 1975 voyage.
- 2018 – 2023** Principal Investigator & Technical Manager in **VIBE** ([VIBE – cradle.buas.nl](https://cradle.buas.nl/)) financed by European Union, OPZuid, Ministry of Economic Affairs, Province of Noord-Brabant, Municipality of Tilburg through a 7 million euro Regional Strategy and Innovation Grant, to develop highly realistic and engaging human face animated avatars based on 3D photogrammetry for medical education.
- 2016-2019** Lead Researcher and Technical Manager in **Maritime Spatial Planning Challenge Platform** (<https://www.mspchallenge.info/>): a multi-awarded European and national funding serious game on Maritime Spatial Planning focused on international collaboration, using simulation based on real data, for the Clyde Estuary, Baltic Sea, North Sea, Adriatic Sea and Eastern Mediterranean.
- 2017-2018** Project Manager in **Smart City Hospitality - Scithos**: European Horizon 2020 research and innovation program to develop a digitally augmented board game, to evaluate policies and key performance indicators of city tourism.
- 2015 – 2018** Lead Researcher and Project Manager in **GURaaS** (<https://guraas.cradle.buas.nl/>): Web based player experience evaluation platform which allows researchers and developers to remotely collect and analyse in-game behaviour.
- 2017** Project Manager in **Dancer Awakens**: First Augmented Reality short movie using the Hololens, show cased in the Go Short - International Short Film Festival of Nijmegen.
- 2016** Lead Researcher and Project Manager in **Green Bubbles Dive**: Horizon 2020 project where an underwater Virtual Reality experience based on a real diving site was developed using photogrammetry.

Invited Public Speaking (with over 40 presenting only the most impactful & relevant)

- 02 June 2026** **Keynote at Logistics Community Brabant - AI Hub**
Topic: EU approach for AI
- 14 Apr 2026** **Invited Speaker at Virtual Productions Gathering 2026**
Topic: AI is making us work hard or work smart?
- 20 Dec 2025** **Invited Speaker Multatuli studentenprogramma 2025**
Topic: EU approach for AI
- 25 Nov 2025** **Panel on Sustainable Technological Futures in Education**
Topic: How Sustainability and AI impacting Cultural Heritage
- 17 Nov 2025** **Invited speaker at Avenue in Universidade Lusofona**
Topic: **Games AI - luck, logic, and white lies**
- 29 Oct 2024** **Panel Studium Generale: Deschooling Society in the Age of AI**
Topic: Deschooling Society in the Age of AI
- 19 Sept 2024** **Panel Mindlabs AI Symposium**
Topic: AI as a Research Assistant
- 10 Sept 2024** **BUas Open of Academic Year 2024: Innovation and Ethical Responsibility**
Topic: Responsible AI and Impacts
- 18 June 2024** **Invited Keynote at KreativEU Summit**
Topic: How Sustainability and AI impacting Cultural Heritage
- 27 May 2024** **Invited Keynote at Sustainable Tourism Mobility Forum**
Topic: Sustainable travel for film productions
- 13 Nov 2023** **Experience immersive rooms at educational institutions yourself | SURF.nl**
Topic: Virtual Productions - Behind the scenes
- 25 Sept 2023** **Invited Speaker at LevelUp**
Topic: Serious markets for serious games
- 10 Aug 2023** **Invited Speaker at Griffith University – Film School**
Topic: Virtual Productions - Behind the scenes
- 6 Aug 2023** **Visiting researcher at Griffith University Australia**
Topic: Participation on New York Counterpoint short film
- 12 May 2023** **Invited Speaker at Werktafel Digitale Zorg.**
Topic: Digital Humans and Serious Game Technology to help to train medical staff
- 11 May 2023** **Invited Speaker at AI Forward Forum and Google Developer**
Topic: AI and Sustainability in Creative Industries
- 10 Mar 2023** **Invited Speaker at Virtual Production Green Symposia**
Topic: AI and Sustainability of new technologies
- 15 Dec 2022** **Invited Speaker at Breda Game City Gathering**
Topic: Games & AI empowering development tools
- 22 Nov 2022** **Cross Media Café, Hilversum, Netherlands**
Topic: Educating Virtual Production Professionals
- 8 Sep 2022** **AI Mini-Summit, Breda, Netherlands**
Topic: Conversational AI
- 14 Apr 2022** **Invited Speaker at Virtual Productions Gathering 2022, Breda, Netherlands**
Topic: XR Lab - Start of the journey
- 23 Mar 2018** **Using games to enrich Maritime Spatial Planning, Texel, Netherlands**
Topic: Applied games and social responsibility
- 22 Nov 2017** **Old Sites New Media, Belmonte, Portugal**
Topic: New Media – Option to improve the Portugal sub-developed regions
- 02 Apr 2016** **Opportunities in Tourism: Inspiration from Serious games, Pernambuco, Brazil**
Topic: Designing Experiences

List of Publications:

- 2026.** (accepted) Carlos Pereira Santos, Phil de Groot, Jens Hagen, Wilco Boode, and Igor Mayer, **Procedural Ocean: Design of Maritime Visualizations using Game Technology**, IEEE Conference on Games, Madrid, Spain
- 2026.** (accepted) Melisa Atish, Carlos Pereira Santos, Carlos, **Usability Evaluation of World Model Studio**, 14th European Conference on Visual Information Processing (EUVIP 2026), Luxembourg
- 2026.** Fonseca, X.; Pereira Santos, C.; Hutchinson, K.; Hagen, J.; Groot, P.D.; Relouw, J.; Mayer, I. **Immersive Ocean: A Virtual Twin for Participatory Decision Support in Maritime Spatial Planning**. *ISPRS Int. J. Geo-Inf.* 2026, *15*, 323. <https://doi.org/10.3390/ijgi15070323>
- 2026.** Perera, T.-M., Hassreiter, S., Haggis-Burridge, M. & Pereira Santos, C., **Avenue : European vocational excellence report 2026**, Open access
- 2026.** Carlos Pereira Santos & Frans Melissen, **Creating a mutually reinforcing relationship between ethical/sustainable AI and Europe's cultural heritage**, Whitepaper, Open access
- 2026.** Boode, W., Nemetsov, L., Santos, C.P. **Mixed Reality Meets Marine Spatial Planning**. in: tom Dieck, M.C., Jung, T., Han, D.I.D., Heller, J. (eds) XR and Metaverse. 2025. Springer Proceedings in Business and Economics. Springer, Cham. https://doi.org/10.1007/978-3-032-11983-4_7
- 2025.** Carlos Pereira Santos, C. & Frans Melissen, **Shaping AI the European Way : BUas leads ethical and sustainable innovation in KreativEU**, Uncover. 2025, 9, p. 26-27
- 2025.** Fonseca, X., Pereira Santos, C., Hutchinson, K., Relouw, J. & Mayer, I., **Towards a virtual twin for the ocean: Technical performance and usability testing of the 'Immersive ocean platform'**. OCEANS 2025 Brest, France
- 2024.** Igor Mayer, M.S. Van Gisbergen, M Haggis-Burridge, Carlos Pereira Santos **Better Futures with Games and Immersive Media**. Evaluation report 2018-2023 for the Research Assessment of R&D@AGM, Open access
- 2023.** Carlos Pereira Santos, Harald Warmelink, M Courchesne. **Digital Twins of the ocean : autogenerated 3D environments for validating offshore wind farm operations**. EventBUILD-IT 2023, Rome, Italy
- 2023.** Carlos Pereira Santos, Phil de Groot, Jens Hagen, A. Boudry, Igor Mayer. **CUBE: Conversational User-Interface-Based Embodiment: developing a digital humans embodiment for conversational agents: design, implementation, and integration challenges**. ACM International Conference on Intelligent Virtual Agents. - Würzburg, Germany (IVA '23)
- 2023.** A. Boudry, Joey Relouw, Alexander van Buggenum, Carlos Pereira Santos, Igor Mayer. **Portraying emotions on digital humans through emotive colour maps**. ACM International Conference on Intelligent Virtual Agents. - Würzburg, Germany (IVA '23)
- 2023.** Carlos Pereira Santos, Joey Relouw, Kevin Hutchinson, Alexander Buggenum, Agathe Boudry, Annemarie Fransen, Myrthe van der Ven, Igor Mayer. **Embodied Agents for Obstetric Simulation Training**. ACM Intelligent User Interfaces (IUI 2023)
- 2022.** Luca Quartesan, Carlos Pereira Santos. **Neural Bidirectional Texture Function Compression and Rendering**. SIGGRAPH Asia 2022
- 2021.** Joey Relouw, M.S. Van Gisbergen, Carlos Pereira Santos. **Recognition of facial expressions in VR: an experiment of still photos versus 3D computer graphic images**. 6th International AR VR Conference
- 2020.** Carlos Pereira Santos, Harald Warmelink, Wilco Boode, Phil de Groot, Kevin Hutchinson, Magali Gonçalves, Jeroen Steenbeek, & Igor Mayer. **A Digital Game-Based Simulation Platform for Integrated Marine Spatial Planning: Design Challenges and Technical Innovations**. The Journal of Ocean Technology, 15(2), 78-100.
- 2019.** Carlos Pereira Santos. **Understanding People through Games** (PhD Thesis) Eindhoven University of Technology Library. ISBN: 978-90-386-4791-3.
- 2019.** Phil de Groot, Wilco Boode, Carlos Pereira Santos, Harald Warmelink, Igor Mayer. **A Shipping Simulation Through Pathfinding: SEL within the MSP Challenge Simulation Platform**. 20th annual European GAME-ON® Conference (GAME-ON®'2019) on Simulation and AI in Computer Games.
- 2019.** Carlos Pereira Santos, Niels van Gaans, Vassilis-Javed Khan, Panos Markopoulos. **Effects of advertisements and questionnaire interruptions on the player experience**. IEEE Conference On Games 2019, UK
- 2019.** Magali Goncalves, Jeroen Steenbeek, Maciej Tomczak, Giovanni Romagnoni, Rikka Puntilla, Ville Karvinen, Carlos Pereira Santos, Xander Keijser, Lodewijk Abspoel, Harald Warmelink, Igor Mayer. **Food-web modeling in the Maritime Spatial Planning Challenge Simulation Platform: Results from the Baltic Sea Region**. International Simulation and Gaming Association. (ISAGA 2019), Warsaw, Poland.

List of Publications (continuation):

- 2019.** Jessika Weber-Sabil, Lidija Lalacic, Thomas Buijtenweg, Kevin Hutchinson, Carlos Pereira Santos, Melissen Frans, Ko Koens and Igor Mayer. **Managing Competing Values in Sustainable Urban Tourism: a Simulation-Gaming Approach.** International Simulation and Gaming Association. (ISAGA 2019), Warsaw, Poland.
- 2019.** Jeroen Steenbeek, Giovanni Romagnoni, Jacob Bentley, Johanna Jacomina Heymans, Natalia Serpetti, Magali Gonçalves, Carlos Pereira Santos, Harald Warmelink, Igor Mayer, Xander Keijser, Rhona Fairgrieve and Lodewijk Abspoel. **Combining ecosystem modelling and serious gaming to aid transnational management of marine space. Managing local and global fisheries.** in Ecology and Society Journal
- 2019.** Carlos Pereira Santos, Kevin Hutchinson, Vassilis-Javed Khan, Panos Markopoulos. **Profiling Personality Traits with Games.** ACM Transactions on Interactive Intelligent Systems Journal (TiiS), Volume 9 Issue 2-3, Article No.: 11, March 2019, ACM New York, NY, USA. ISSN:2160-6455 EISSN:2160-6463, doi: 10.1145/3230738
- 2018.** Carlos Pereira Santos, Vassilis-Javed Khan & Panos Markopoulos. **Profiling ethics orientation through play.** Behaviour & Information Technology Volume 37, 2018 - Issue 9: Highlights from ECCE 2018, September 2018
- 2018.** Carlos Pereira Santos, Vassilis-Javed Khan, Panos Markopoulos. **Interactive Narratives for Profiling Ethics Orientation.** 36th European Conference on Cognitive Ergonomics (ECCE 2018), September 2018, the Netherlands
- 2018.** Marnix Van Gisbergen, Hans Lodders, Carlos Pereira Santos, Dennis Hoogervorst. **How AR visualisation can help us understand data.** Data Driven Digital Marketing Event (DDME2018), 2018, Huizen, Netherlands.
- 2018.** Kevin Hutchinson, Harald Warmelink, Wilco Boode, Carlos Pereira Santos, Igor Mayer. **An Offshore Energy Simulation through Flow Networks: CEL within the MSP Challenge 2050 Simulation Game Platform.** European Simulation and Modelling Conference (ESM 2018), October 2018, Belgium
- 2017.** Carlos Pereira Santos, Jeroen van de Haterd, Kevin Hutchinson, Vassilis-Javed Khan, Panos Markopoulos. **GURaaS: An End-User Platform for Embedding Research Instruments into Games.** 6th International Symposium on End User Development (IS-EUD 2017), the Netherlands
- 2017.** Carlos Pereira Santos, Kevin Hutchinson, Vassilis-Javed Khan, Panos Markopoulos. **Measuring Self-Esteem with Games.** ACM Intelligent User Interfaces (IUI 2017), March 2017, Cyprus
- 2017.** Licia Calvi, Carlos Santos, Joey Relouw, Bojan Endrovski, Serena Lucrezi, Antonio Sarà, Ubaldo Pantaleo. **A VR game to teach underwater sustainability.** The Fifth IFIP Conference on Sustainable Internet and ICT for Sustainability (SustainIT 2017), December 2017, Madeira, Portugal.
- 2017.** Licia Calvi, Carlos Pereira Santos, Serena Lucrezi, Antonio Sarà, Joey Relouw, Bojan Endrovski, Chris Rothwell, Marco Palma, Ubaldo Pantaleo. **Can a VR game increase awareness in a sustainable marine behaviour?** European Marine Science Educators Association Conference (ESMEA 2017), October 2017, Valletta, Malta.
- 2016.** Carlos Pereira Santos, Vassilis-Javed Khan, Panos Markopoulos. **Game User Research as a Service.** ACM CHI Conference on Human Factors in Computing Systems 2016 Workshop (CHI 2016): Lightweight Games User Research for Indies and Non-Profit Organizations, May 2016, USA
- 2016.** Carlos Pereira Santos, Vassilis-Javed Khan, Panos Markopoulos. **Inferring A Player's Need For Cognition From Hints.** ACM Intelligent User Interfaces (IUI 2016), March 2016, USA
- 2016.** Erwin Peters, Bram Heijligers, Josse de Kievith, Xavier Razafindrakoto, Ruben Van Oosterhout, Carlos Pereira Santos, Igor Mayer, Max M Louwerse. **Design for Collaboration in Mixed Reality Technical Challenges and Solutions.** 8th International Conference on Virtual Worlds and Games for Serious Applications (VS-games 2016), November 2016, Spain
- 2014.** Carlos Pereira Santos, Vassilis-Javed Khan, Panos Markopoulos. **On Utilizing Player Models to Predict Behavior in Crowdsourcing Tasks.** International Conference on Social Informatics (SocInfo 2014), November 2014, Spain
- 2014.** Giuseppe Maggiore, Carlos Santos, Aske Laat. **Smarter smartphones: understanding and predicting user habits from GPS sensor data.** Procedia Computer Science Volume 34, 2014
- 2013.** Giuseppe Maggiore, Carlos Santos, Dino Dini, Frank Peters, Hans Bouwknegt, Pieter Spronck. **LGOAP: adaptive layered planning for real-time videogames.** IEEE Conference on Computational Intelligence in Games (CIG2013), August 2013, Canada
- 2005.** José Danado, Eduardo Dias, Teresa Romão, Nuno Correia, Adelaide Trabuco, Carlos Santos, João Serpa, Manuel Costa, António Câmara. **Mobile Environmental Visualization.** The Cartographic Journal Vol. 42 No. 1, June 2005
- 2004.** Teresa Romão, Nuno Correia, Eduardo Dias, Jose Danado, Adelaide Trabuco, Carlos Santos, Rossana Santos, Antonio Camara, Edmundo Nobre. **ANTS - Augmented Environments.** Computers & Graphics, Special Issue on Pervasive Computing and Ambient Intelligence - Mobility, Ubiquity and Wearables Get Together, October 2004

List of Publications (continuation):

- 2004.** Miguel Boavida, Anacleto Correia, Susana Cabaço, Adelaide Trabuço, Carlos Santos and Jorge Pereiros. **Tools for Experiential Learning: The "E2-LEARNING" Approach.** International Conference on Education (IADAT-e2004) Volume: IADAT-e Technological Issues, July 2004, Spain
- 2004.** José Danado, Eduardo Dias, Teresa Romão, Nuno Correia, Adelaide Trabuço, Carlos Santos, João Serpa, Manuel Costa, António Câmara. **A Multi-user mobile system to visualize environmental processes.** Proceedings of the 12th International conference on Geoinformatics-Geospatial Information Research: Bridging the Pacific and Atlantic, June 2004, Sweden
- 2003.** José Danado, Eduardo Dias, Teresa Romão, Nuno Correia, Adelaide Trabuço, Carlos Santos, David Araújo, Pedro Duarte, Rui Rebocho, José Palmeiro, João Serpa, Manuel Costa, António Câmara. **Mobile Augmented Reality for Environmental Management (MARE).** Eurographics 2003 (EG'03), September 2003, Spain
- 2003.** Carlos Santos, Nuno Correia. **A Navigation and Registration System for Mobile and Augmented Environments.** Workshop on the Design Specification and Verification of Interactive Systems (DSV-IS 2003), June 2003, Portugal.
- 2003.** Miguel Boavida, Rossana Santos, Carlos Pereira Santos, Nuno Correia. **A Framework for Shared and Personalized Video Based Documents.** The Twelfth International World Wide Web Conference (WWW 2003 Posters), May 2003, Hungary
- 2003.** Nuno Correia, Teresa Romao, Carlos Santos, Adelaide Trabuço, Rossana Santos, Luis Romero, Jose Danado, Eduardo Dias, Antonio Camara, Edmundo Nobre. **A Flexible Augmented Reality Architecture Applied to Environmental Management,** SPIE Electronic Imaging, January 2003, USA
- 2002.** Teresa Romão, Eduardo Dias, José Danado, Nuno Correia, Adelaide Trabuço, Carlos Santos, Rossana Santos, Edmundo Nobre, António Câmara, Luís Romero. **Augmenting Reality with Geo-Referenced Information for Environmental Management.** Proceedings of the 10th ACM international symposium on Advances in geographic information systems (GIS'02), November 2002, USA

Supervision of PhD and Professional Doctorate

- Since 2024. Wilco Boode (ongoing).** Daily supervisor of PhD with Eindhoven University of Technology, Netherlands
Topic: Design of shared Mixed Reality experiences
- Since 2024. Jonas Pastor (ongoing).** External advisor of Professional Doctorate with BUAs, Netherlands
Topic: Design of shared Mixed Reality experiences

Supervision of Master Thesis

- 2023. Dana van der Wal.** Master Game Technology. Thesis entitled:
From Pixels to Partners: Enhancing Digital Pet Interaction through Adaptive Learning
- 2022. Luca Quartesan.** Master Game Technology. Thesis entitled:
Neural Bidirectional Texture Function Compression and Rendering
- 2021. Andrei-Ion Onița.** Master Game Technology. Thesis entitled:
How do players relate the number of training steps with the skill of a ML NPC agent?
- 2021. Jens Hagen.** Master Game Technology. Thesis entitled:
Agent Participation in First Person Shooter Games using Reinforcement Learning and Behaviour Cloning
- 2021. Dyon Kreffer.** Master Game Technology. Thesis entitled:
Reinforcement learning for non-player characters in the video game industry
- 2020. Thomas Goussaert.** Master Game Technology. Thesis entitled:
DNIIK – Data Driven Full Body Inverse Kinematics for Virtual Reality
- 2019. Heleen van Geelen.** Master Media Innovation. Thesis entitled:
The Video Game Industry: Qualitative Insights in Player Experience Evaluation and Data Collection.
- 2019. Niels van Gaans.** Master Media Innovation. Thesis entitled:
The effect of cognitive load demanded by in-game questionnaires on game experience
- 2018. Julie Jurgens.** Master Media Innovation. Thesis entitled:
Home Interior goes 360 The contribution of a 360-degree Social Network Services to home interior design
- 2018. Yonathan Klijnsma.** Master Game Technology. Thesis entitled:
An anti-cheat protection engine for Windows based games

Supervision of Master Thesis (continuation)

- 2017. Hans Lodders.** Master Media Innovation. Thesis entitled:
How Augmented Reality supports Media Strategies by improving brand segmentation through 3D data visualization
- 2017. Nan Chen.** Master Media Innovation. Thesis entitled:
Virtual Reality enabled shopping and the business potential: A case study
- 2017. Hicham Filali.** Master Media Innovation. Thesis entitled:
Most important features influencing the mobile retail shopping experience
- 2017. Jordy Ruiter.** Master Game Technology. Thesis entitled:
Project Kubed: Maintaining online platforms using cloud container systems for continuous integrations and deployment.
- 2016. Kevin Hutchinson.** Master Game Technology. Thesis entitled:
Modelling player self-esteem from in-game behavior.
- 2016. Bart van der Kruys.** Master Game Technology. Thesis entitled:
Utility AI: An Autonomous AI Tool
- 2016. Ajay Ramautar.** Master Game Technology. Thesis entitled:
Hybrid / combined behaviour system in games
- 2016. Danny A. Francken.** Master Media Innovation. Thesis entitled:
RSPND: Connect Digital to Physical; let the products inform you.
- 2015. Pieter Vader.** Master Media Innovation. Thesis entitled:
Desktop or Head Mounted Displays: The role of the displayed environment in the effect on Presence, Engagement, Naturalness and Visit Intention in a virtual museum
- 2015. Vincent Zuidema.** Master Media Innovation. Thesis entitled:
WeBoat: Opportunities for peer to peer boat sharing
- 2015. Yvonne Kolen.** Master Media Innovation. Thesis entitled:
Interactive Digital Narratives and the Narrative Paradox: A user modeling approach to reduce the narrative paradox.
- 2014. Kenny Wood.** Master Media Innovation. Thesis entitled:
Exploring User Acceptance of Virtual Reality Technologies to Visualize indoor event venue floorplan layouts.
- 2014. Megan Wang.** Master Media Innovation. Thesis entitled:
The context aware connecting students through social media and games
- 2013. Matt Merry.** Master Media Innovation. Thesis entitled:
Public Acceptance of Digital Flyer systems
- 2012. Pieter Wels.** Master Media Innovation. Thesis entitled:
Design for Design: Playful Creation Processes to improve 3d Print Product Design

Supervision Graduations Projects (only recent ones)

- 2026. Jonathan van Santen (ongoing).** Applied Data Science and Artificial Intelligence. Bachelor graduation project:
HEIMDALL – Human-centric Ethical Intelligence Monitoring Data Access and Legal Languages
- 2026. Melisa Atish.** Creative Technologies and Artificial Intelligence (Howest, Belgium). Bachelor graduation project:
DREAMWORLD - Dynamic Rendering EnAbleD Models World
- 2025. Andrea Tosheva.** Applied Data Science and Artificial Intelligence. Bachelor graduation project:
Sentiment analysis of based transcription text of qualitative research interviews
- 2025. Hidde Derks.** Creative Media and Gamer Technologies. Bachelor graduation project:
Continuous integration and automatic deployment tools for games
- 2024. Bas Walhout.** Creative Media and Gamer Technologies. Bachelor graduation project:
Energy consumption monitoring and control of XR Stage
- 2023. Merlijn van der Kamp.** Creative Media and Gamer Technologies. Bachelor graduation project:
Web-marbles: a low-maintenance-cost open source multiplayer game using 3D web technologies
- 2023. Tuan Le.** Creative Media and Gamer Technologies. Bachelor graduation project:
Tools Programming for Player versus Player mobile game slot heroes: case study.
- 2022. Michiel Pater.** Creative Media and Gamer Technologies. Bachelor graduation project:
Network platform to support real time multiplayer mobile games